THOTA PASS

One update before they leave home

From a morning at the roadside to a working prototype and a business plan for coffee estates in Karnataka, India.

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Live app: https://thota-pass-harvest-pickup.exoticbay3.chatgpt.site/app/ Estate office PIN 4286 · Driver PIN 7314

Code and videos: https://github.com/SampadaDd/thota-pass

At a glance	
The problem	Coffee pickers walk to a roadside junction at dawn without knowing if the van is coming. The one person who knows, the coordinator, is driving it.
The insight	It is not a labour shortage. It is information that changes between night and morning, held by someone who cannot answer the phone.
The product	The driver taps one status before he drives. Every worker gets it by app, a Kannada text and automated call, or a radio buzzer at home.
The evidence	Five interviews. Two usability rounds (the first failed; the fix worked). Drivers and owners in Karnataka shown the prototype; Coffee Board feedback.
The business	The estate owner pays: ₹5 per worker per day or a ₹2,999 Season Pass. Next: a picking calculator that gives owners a reason to pay.

1. The problem

> *“Nobody knew if the vehicle was coming.”*

Lokesh picks coffee in the hills of Karnataka. One harvest morning he packed his food, left home at 6:30 and walked to the junction where the milk van stops. Seven o’clock passed. Seven-thirty passed. No van, and no news. He called the coordinator, who did not pick up. He reached the estate at 8:30.

He was still paid for the day. What he lost was his morning and a little more trust. The next time the coordinator called, he asked first: “Vehicle confirmed, right?” And he still went, because, in his words, “we need work.”

Why it matters

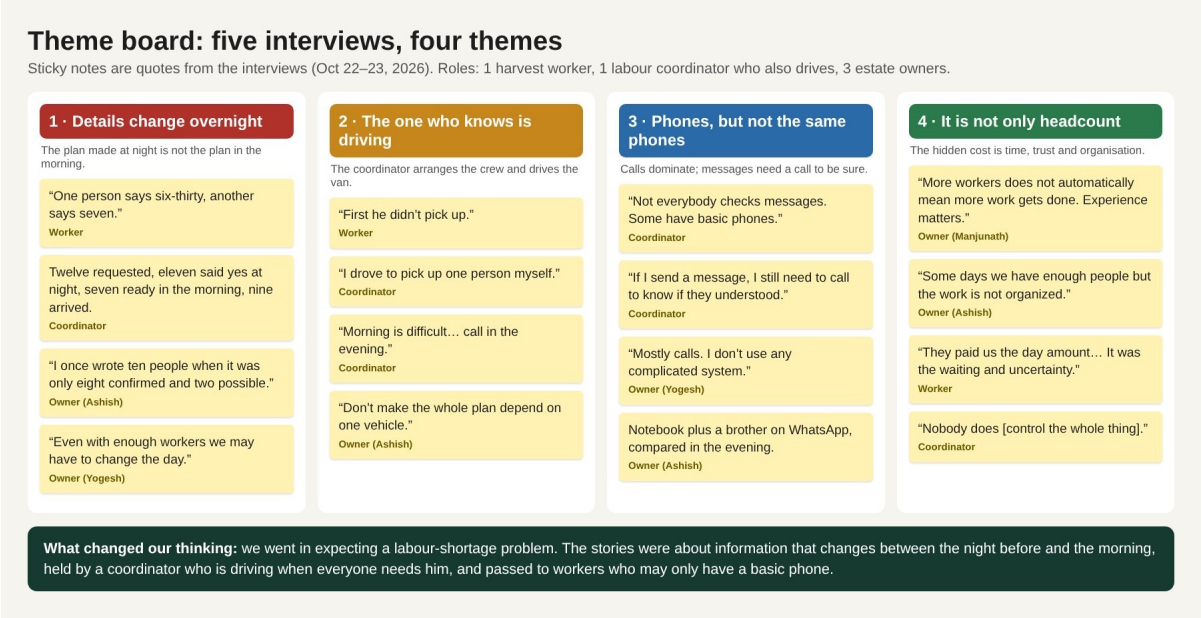
Coffee sustains the livelihoods of over two million people in India. Nearly 99% of holdings are small farms, which grow about 70% of the crop, and Karnataka produces the largest share (Press Information Bureau, 2025). Picking from November to March is done by hand. Estates in Kodagu, Chikkamagaluru and Hassan have reported labour shortages for about five years, rely more on crews who travel from farther away, and small planters say transport costs eat their margins (The Hindu, 2025).

Phones are common but uneven: 85.5% of Indian households own at least one smartphone (MoSPI, 2025), yet a household phone is not the picker’s own phone at 6 a.m., and in the hills the signal itself can fail. Today’s tools are a phone call the night before, WhatsApp for those who read it, and a paper notebook. None of them tells a worker, before he leaves home, that the van is late. I found no count of how many workers lose mornings this way, so I do not claim one.

2. What I learned from people

I interviewed five people in Kannada and English, with consent: a harvest worker, a labour coordinator who also drives the van, and three estate owners. I went in expecting a labour shortage. The stories pointed somewhere else.

Finding	What people said
The plan made at night is not the plan in the morning	An owner asked for **12** workers. **11** said yes at night. **7** were ready in the morning. **9** arrived. “One person says six-thirty, another says seven.”
The person who knows is behind the wheel	“I drove to pick up one person myself.” At 6 a.m. the coordinator cannot answer: “Morning is difficult… call in the evening.”
Not everyone has the same phone	“Not everybody checks messages. Some have basic phones. And if I send a message, I still need to call to know if they understood.”
It is not only headcount	“More workers does not automatically mean more work gets done.” “We should be careful blaming everything on labor shortage.”



Theme board: quotes from all five interviews, clustered into four themes.

Point of view

A seasonal coffee worker in rural Karnataka needs to confirm the pickup and work details **before he leaves home**, because unreliable transport and mixed messages waste his morning and make distant jobs less worthwhile, even when his daily wage is paid.

My first draft said late vans cost workers their wages. The transcript said they were paid. Correcting that changed the whole design: the goal became saving the morning and the trust, not the wage.

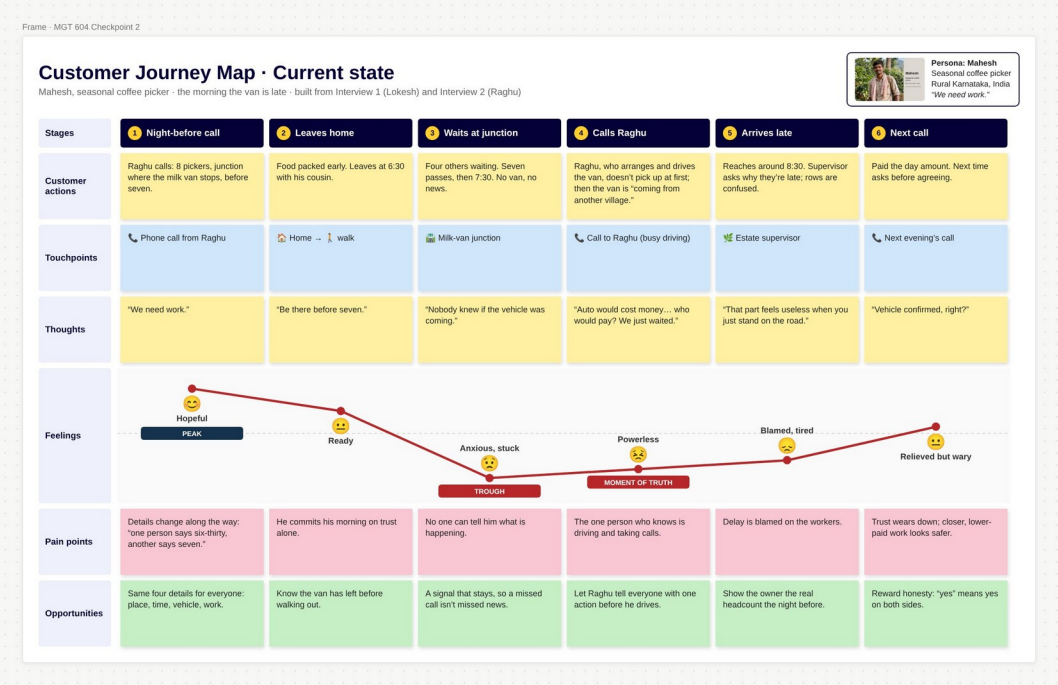
3. From 18 ideas to one tap

I generated 18 ideas with six techniques: a solo brainstorm on paper first, AI-assisted ideation with the constraint "use only the phones people already have", an AI mash-up with a beehive, an airline-loyalty analogy, a team brainstorm and storyboards.

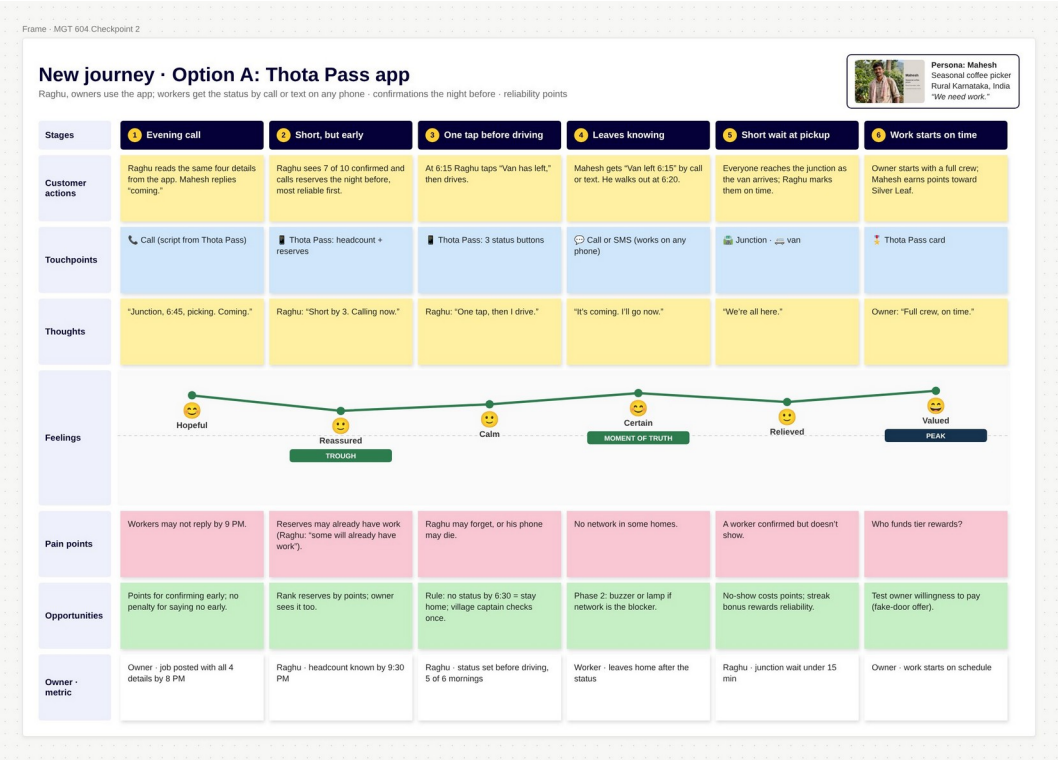
Top three	Desirability	Feasibility	Viability
Village captain's flag	Uses only what people have	Needs a volunteer every morning; cannot say "delayed until 7:05"	Free, but nobody to charge
Long-range radio buzzer	Works with no phone or signal	Hardware to buy, charge and maintain	₹1,499 per buzzer; costly at scale
Thota Pass (chosen)	Fixes the moment of truth before he leaves home	A web app with texts and calls; depends on the driver tapping	Low running cost; the owner pays

The flag was my first favourite, born from the beehive's "a signal that stays until it changes." I changed direction when I saw that it adds one more person to a chain where, as the coordinator put it, "nobody" controls the whole thing. I kept its lesson: whatever I built had to work without a smartphone. The buzzer became the answer for homes with no signal.

4. The solution



Before: the low point is the roadside at 7:30, and the moment of truth is an unanswered call.

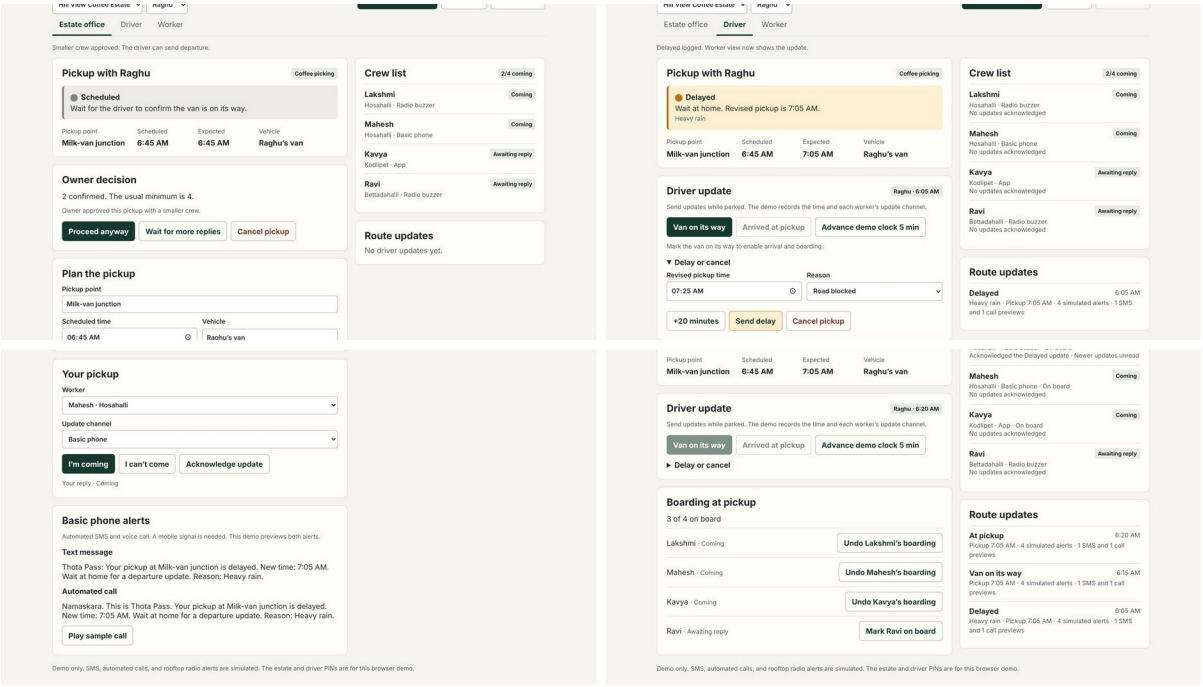


After: the driver taps once at 6:15, and the moment of truth moves to Mahesh's home.

What the prototype does

- **Estate office:** plans the pickup with four details (place, time, vehicle, work). Sees who confirmed. If the crew is short, decides the night before: proceed, wait or cancel.

- **Driver:** while parked, taps one status: on its way, delayed (with a reason and a new time) or cancelled. Then marks arrival and checks workers in. Every update is logged with its time.
- **Worker:** sees the status in plain words, “Delayed. Wait at home. Revised pickup is 7:05 AM.” On a basic phone, the same update arrives as a Kannada text and an automated call that begins “Namaskara. This is Thota Pass.”
- **Thota Buzzer** (concept): for homes with no signal, a radio buzzer glows green, amber or red until the driver changes it.



Estate office, driver delay, basic-phone alerts and the boarding list. Texts, calls and buzzer alerts are simulated in the prototype.

Watch it

The repository holds four videos: the app demo in English (4:15) and Kannada (6:29), and the Thota Buzzer story in English (2:00) and Kannada (3:01).

5. Putting it in people’s hands

Test	What I measured	Result	What it means
Usability, Round 1 (classmate)	Six tasks across office, driver and worker	**4 of 6** completed. Picking up the crew failed	Rejected. The causes were my design: a greyed-out button with no hint, a buried boarding list
Usability, Round 2 (after fixes)	Same tasks on the fixed app	All six completed. Boarding took **5 s**, ease **7/7**	Inconclusive: one tester, one rating missing
Drivers in Karnataka (H1)	Willingness to send an update before driving	Willing to try and learn it, with a cash incentive	Intent, not yet behaviour. Pass mark: 12 of 15 mornings

Test	What I measured	Result	What it means
Estate owners (H2)	Willingness to pay	Mixed: some would pay, some had questions first	Open. Pass mark: 2 of 5 pay a ₹500 refundable deposit
Coffee Board president	Expert view	“A very good approach”; owners need more to buy	Shaped the next version

The moment that mattered most came in Round 2. When the delay arrived, the tester said: “I would stay back home… I would wait until that time.” That is the exact behaviour the product exists for. The miss: after acknowledging, “I cannot see any confirmation,” so the receipt needs to be bigger.

I am careful about what this proves. Friendly testers on laptops and a “yes” in conversation are early signals, not evidence that a driver will tap at 6 a.m. That is what the pilot is for.

6. Business model

Who pays, and why

The worker feels the pain, but the **estate owner pays**. Workers and drivers are never charged. The owner gets a crew that arrives on time, an early warning when the crew is short, and a record of what happened. The driver earns points for timely updates, which the estate pays out at the end of harvest.

Revenue stream	Price	For whom
Pay as you go	₹5 per worker per day	Small growers; pay only on days they hire
Season Pass · Small	₹2,999 per harvest	Up to 15 workers, unlimited jobs, driver points included
Season Pass · Medium	₹6,999 per harvest	Up to 40 workers, several crews
Thota Buzzer	₹1,499 per buzzer, one-time	Homes with no signal
Owner tools (next)	Premium tier, to be priced	Picking calculator and estate notes

A two-week free trial lowers the risk of trying something new, which matters because owners told me a new product comes with questions first.

Costs and unit economics

The main running cost is texts and automated calls to workers on basic phones, which I estimate at **₹1.5–2 per worker per day** from provider rates (not yet a quote). Workers on the app cost almost nothing. At ₹5 per worker per day, that leaves roughly ₹3 per worker-day to cover hosting, support and growth.

An illustration, not a forecast: a small estate with 15 workers, all on basic phones, picking for 60 days would generate 900 worker-days. Messaging would cost about ₹1,350–1,800, against ₹2,999 for the Season Pass or ₹4,500 at pay-as-you-go. Other costs are hosting, field onboarding in Kannada, and buzzer hardware with long-range radio relays (about ₹13,000–20,000 per village group, my estimate).

Channels

- **The Coffee Board**, where growers meet to learn about new plants, soil and techniques. Its president has seen the prototype and gave feedback.
- **Coordinators and drivers**, who already hold the phone numbers and relationships.
- **Word of mouth between neighbouring estates**, which already share vehicles and workers.

Risks

- **Drivers don't tap at 6 a.m.** Fallback: the estate office or a village captain sends the update.
- **Owners won't pay for pickup alone.** Response: bundle the owner tools below; test pay-as-you-go.
- **No signal in some homes.** Response: the buzzer, built only if the pilot shows signal is the real blocker.

7. What's next

Give owners a reason to pay

For the course I deliberately solved one problem. The Coffee Board's feedback made the next step clear: owners need tools for their own work.

- **Picking calculator.** Each worker's sacks are counted every day, at 50 kg a sack, and paid weekly, bi-weekly, at the end of the season, or early as an advance. Owners could use Excel, but most find it too complicated. The app would keep the count and the totals for every payment schedule.
- **Estate notes.** Blocks, rates and techniques shared in one place, so the knowledge owners invest in is not lost.

Harvest-season pilot

When	What	Pass mark
December, week 1	Set up one estate near Sakleshpur, through my own contacts or an introduction through the Coffee Board. One route, 3–5 drivers, workers' numbers with consent.	Ready to run
December	H1, Wizard of Oz: each driver sends one message before driving; I relay a Kannada text and call to his workers.	12 of 15 driver-mornings; junction wait under 15 minutes (about an hour today)
January	H2: show owners the version with the picking calculator and ask for a refundable deposit.	2 of 5 owners pay ₹500 toward the ₹2,999 Season Pass
January 31	Decide: launch for next season, move updates to the estate office, or change the price.	Decision based on both results

From prototype to product

- A shared database and a real text and voice provider (today data stays in one browser and alerts are simulated).
- A full Kannada interface and a clearer acknowledgement receipt.

- A first Thota Buzzer unit, only if the pilot shows signal is the blocker.

8. What this project taught me

- **Interviews can overrule the brief.** I was looking for a labour shortage and found a coordination failure.
- **Check your own claims.** My first point of view about lost wages was wrong; the transcript corrected it.
- **Test the test.** Round 1 failed partly because I never walked through my own worksheet.
- **Can is not will.** A friendly test shows that people can use something, not that they will when no one is watching.
- **The worker feels the pain, but the owner pays.** Solve one problem well, then add what the payer needs.

My role

I led the project end to end: interviews in Kannada and English, synthesis, persona and journey maps, ideation, the prototype, the hypotheses and test plan, two usability rounds, the conversations with drivers, owners and the Coffee Board, and the demo videos in two languages. I used Claude and ChatGPT for translation, ideation, building and editing, and checked every quote against the transcripts.

Sources

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Devaiah, D. B. P. (2025, January 17). Big coffee planters in Karnataka see relief, but small planters continue to face labour shortage. The Hindu, via Kodagu First.

Ministry of Statistics and Programme Implementation (2025). Comprehensive Modular Survey: Telecom, 2025.

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Mahesh is a composite persona built from the interviews. Interviews were recorded with consent.